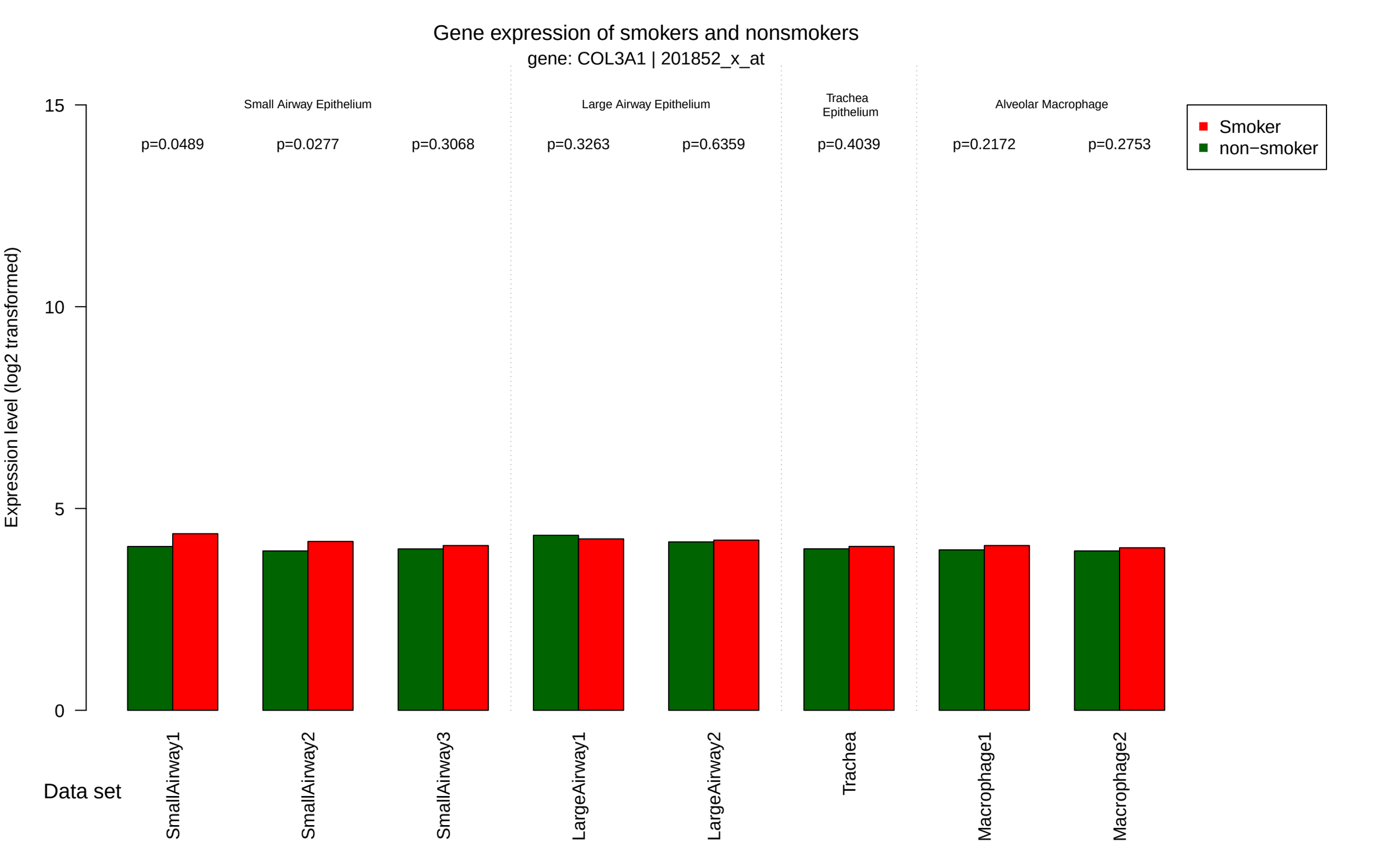
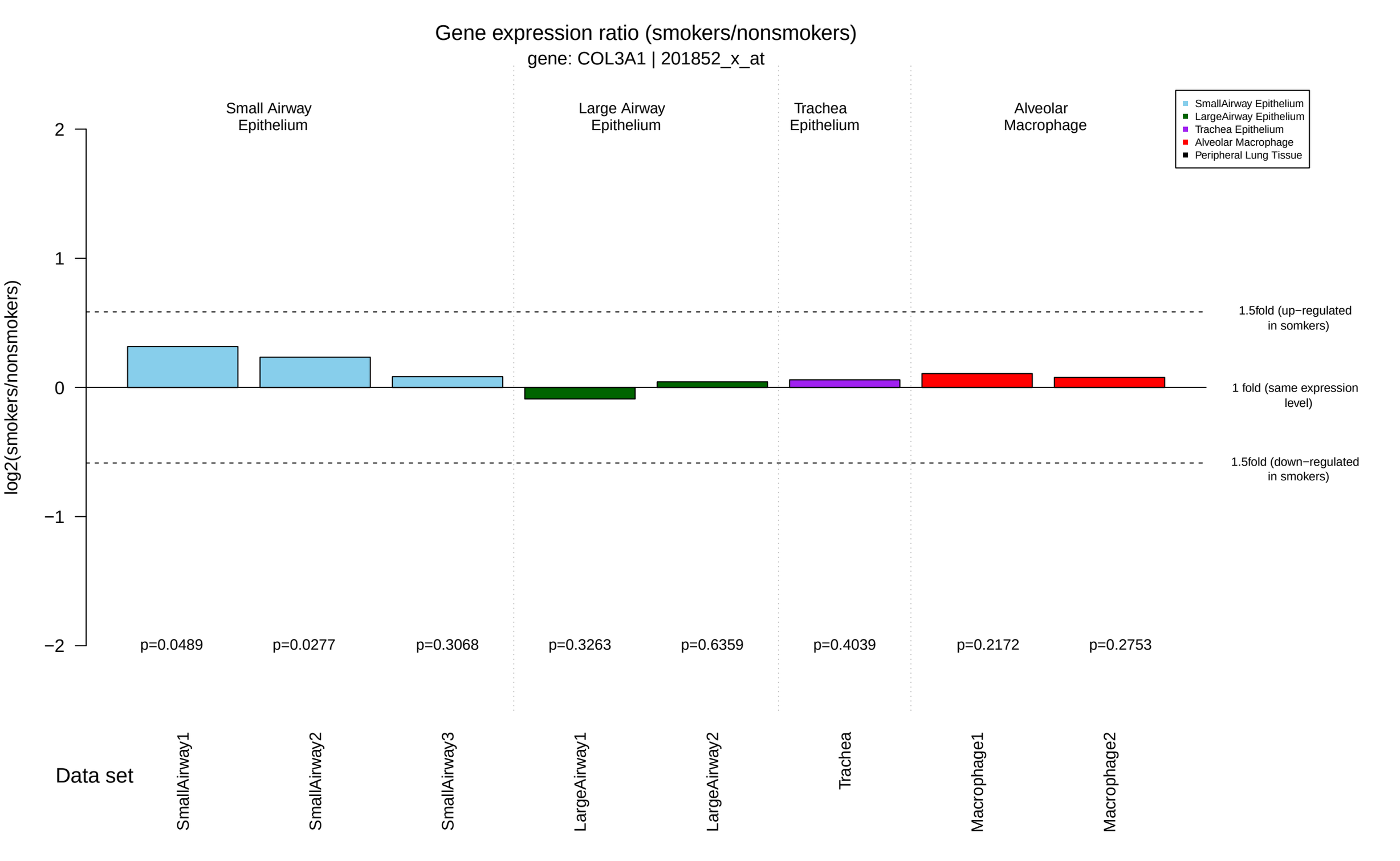
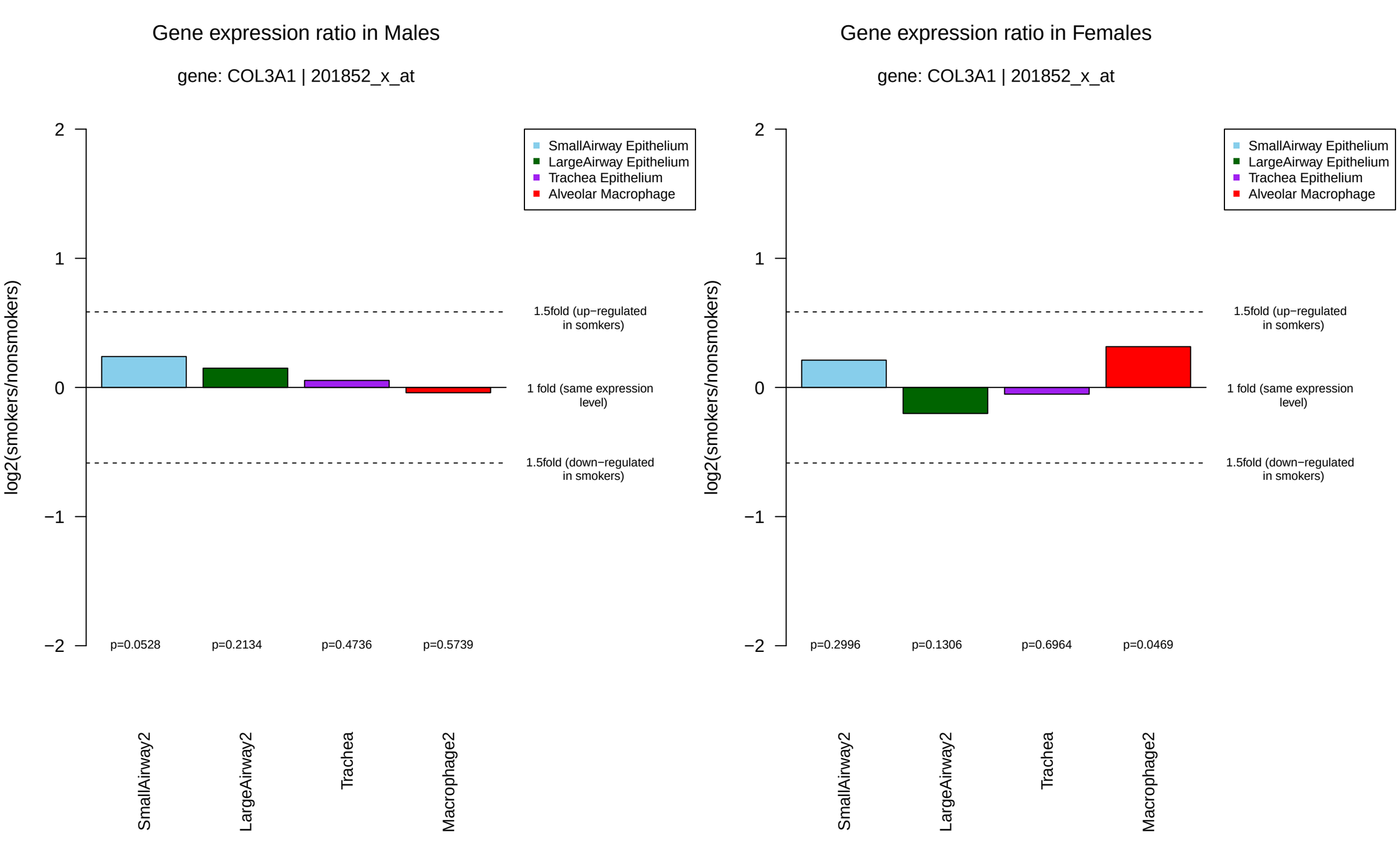
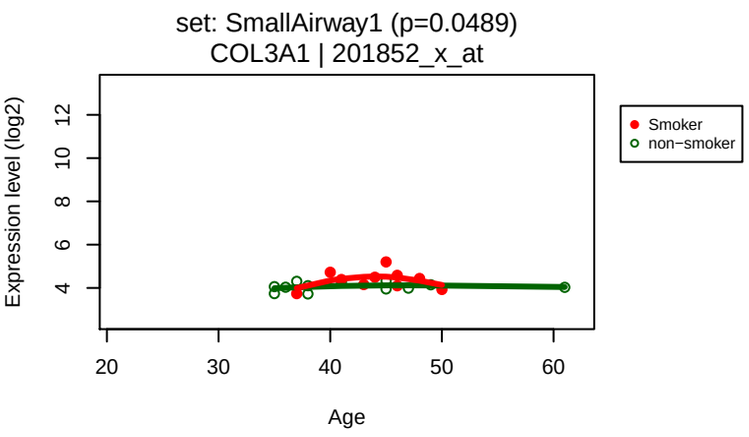
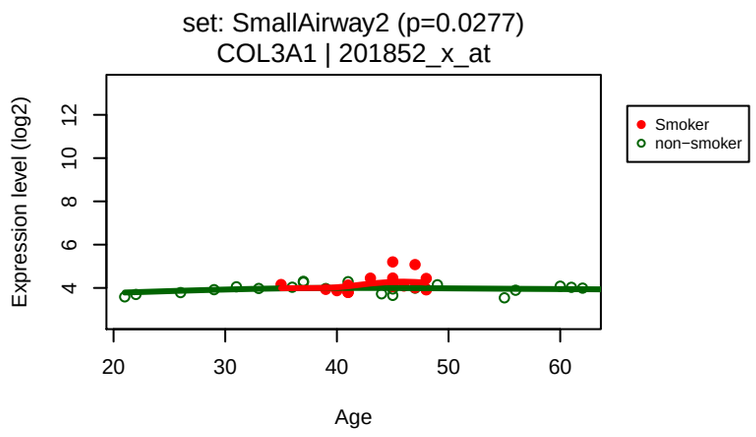
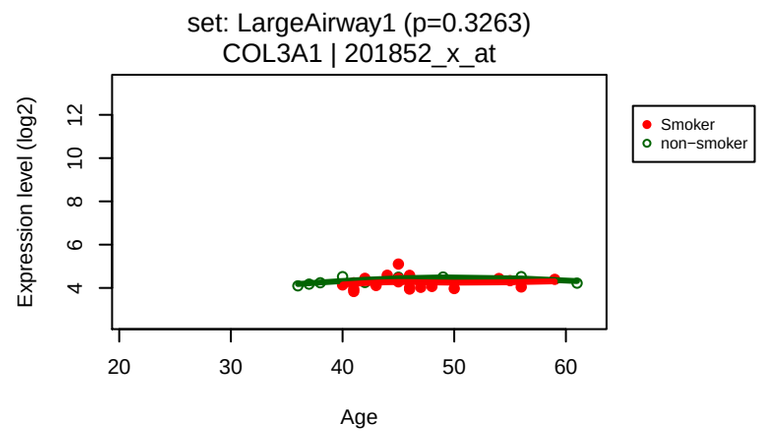

Gene expression of smokers and nonsmokers
gene: COL3A1 | 201852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0489
p=0.0277
p=0.3068
p=0.3263
p=0.6359
p=0.4039
p=0.2172
p=0.2753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

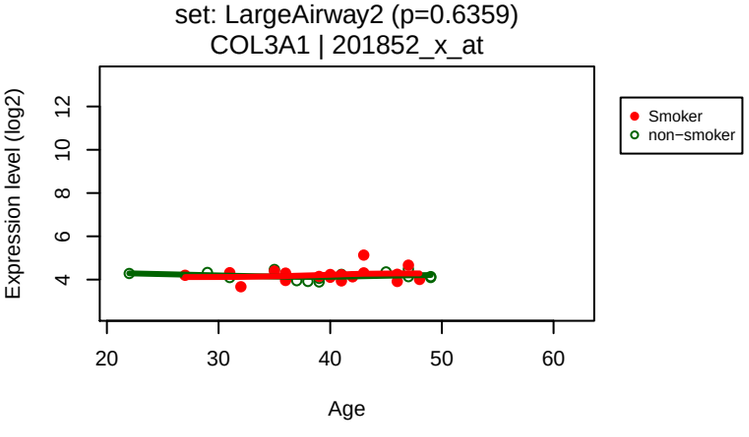

Gene expression ratio (smokers/nonsmokers)
gene: COL3A1 | 201852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0489
p=0.0277
p=0.3068
p=0.3263
p=0.6359
p=0.4039
p=0.2172
p=0.2753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

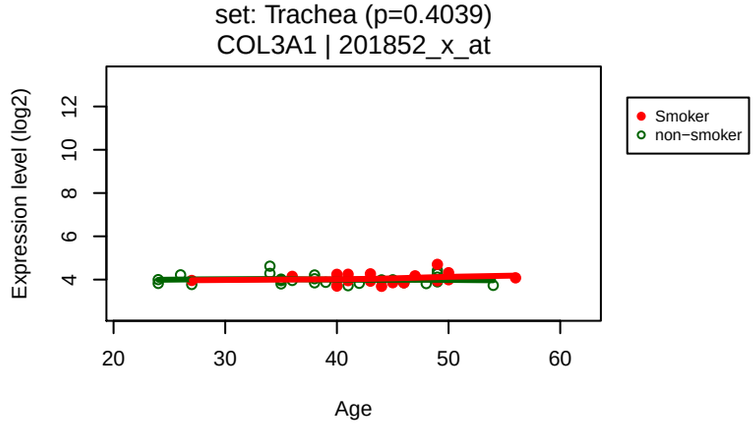

Gene expression ratio in Males
gene: COL3A1 | 201852_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0528
p=0.2134
p=0.4736
p=0.5739
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL3A1 | 201852_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2996
p=0.1306
p=0.6964
p=0.0469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

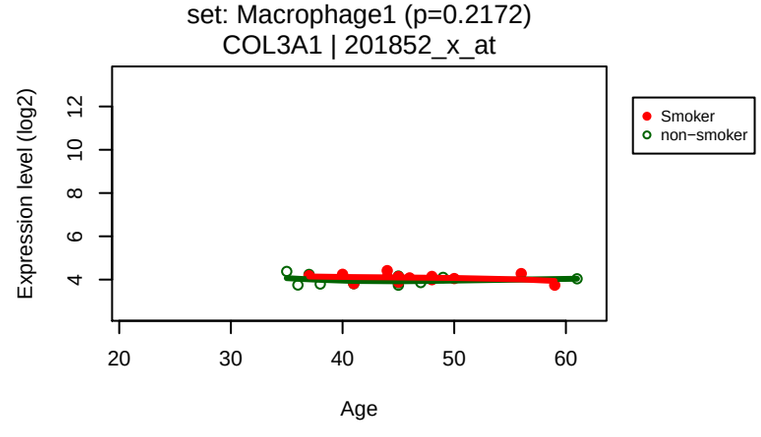

set: SmallAirway1 (p=0.0489)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker

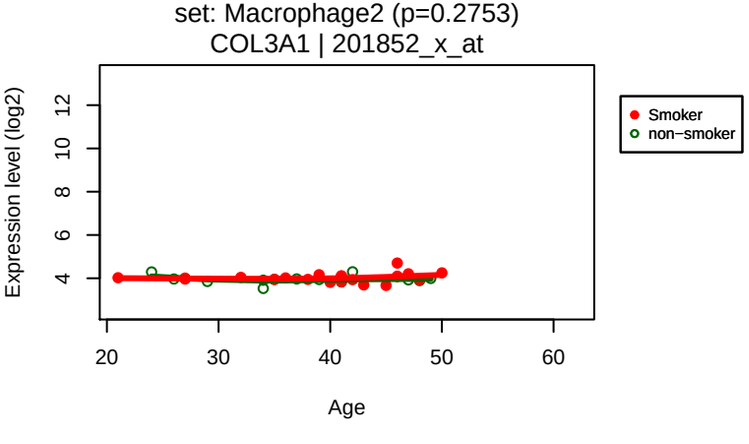

set: SmallAirway2 (p=0.0277)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker

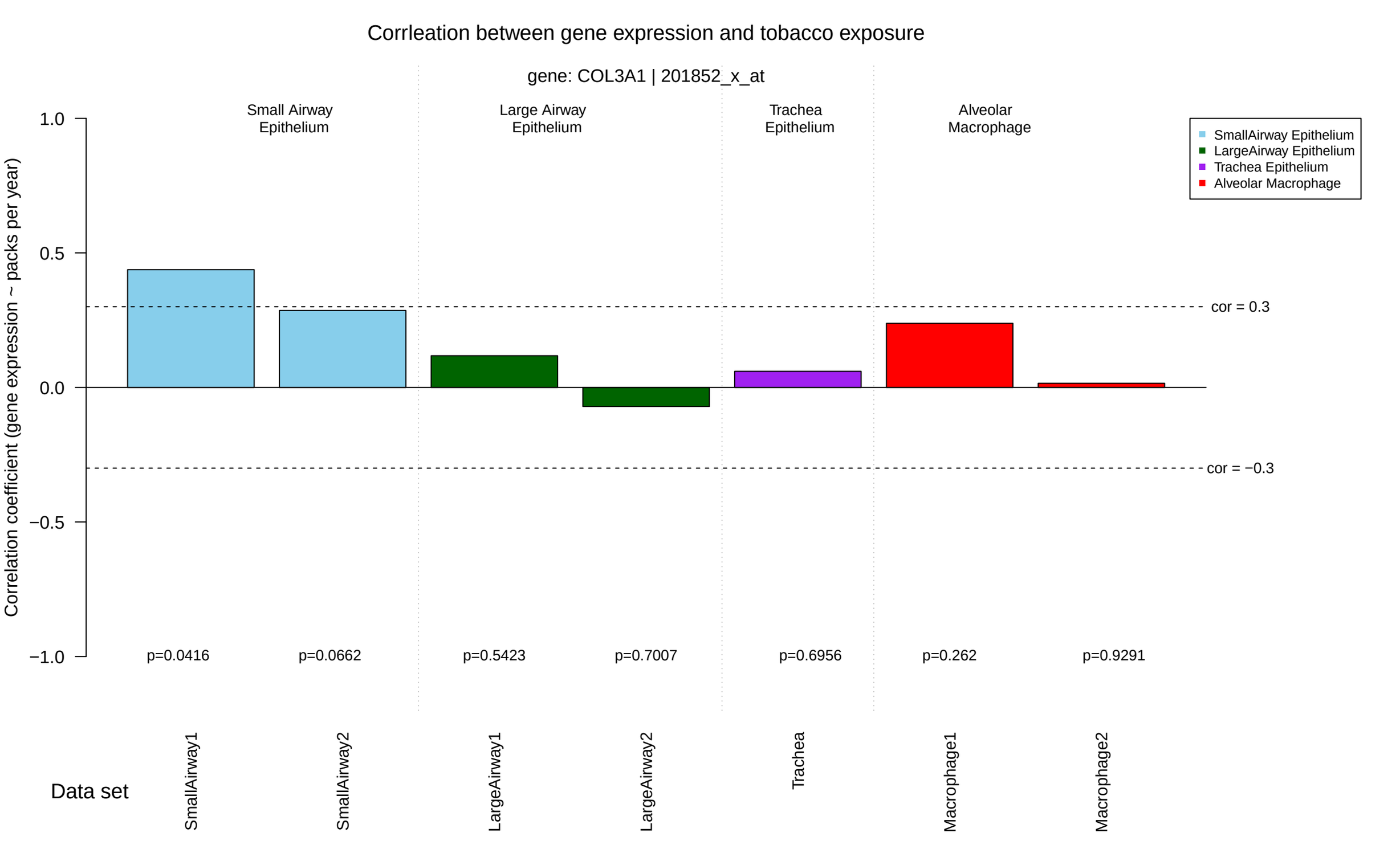

set: LargeAirway1 (p=0.3263)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6359)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4039)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2172)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2753)
COL3A1 | 201852_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL3A1 | 201852_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0416
p=0.0662
p=0.5423
p=0.7007
p=0.6956
p=0.262
p=0.9291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2